Pranav Kalakota

West Lafayette, IN | pkalakot@purdue.edu | linkedin.com/in/pranav-kalakota | github.com/Pranavkalakota

Education

Purdue University West Lafayette, IN
Bachelor of Science in Computer Science Aug. 2025 – May 2029

- Coursework: Programming in C, Object-Oriented Programming, Linear Algebra, Discrete Mathematics, Statistical Methods

Experience

Software Engineering Intern June 2026 – Present
Crcle *West Lafayette, IN*

- Architected a multimodal RAG pipeline fusing Whisper ASR, Llama 3.2 3B, LLaVA 7B, ArcFace, and ECAPA-TDNN with cosine-similarity gating over ChromaDB ONNX vector store, enabling zero-touch HCI via cross-app voice control with sub-20s latency and zero cloud dependency
- Engineered a schema-constrained NLP intent router with 3-tier fuzzy/regex/LLM fallback and semantic vector search, achieving 95% deterministic routing across 500+ fuzz invariants and 12+ query types in production
- Optimized always-on wearable ASR pipeline with adaptive windowing and CPU-routed Whisper inference, cutting transcription latency 50% (12s to 6s) and wake-word activation 90% (2s to <200ms) via offline-first ONNX detection with parallel ASR fallback
- Built a cross-platform vision-based GUI automation system achieving sub-2s latency and 90%+ element localization accuracy across arbitrary applications by routing screenshots through Gemini multimodal LLM inference with pixel-precise coordinate extraction and exponential-backoff retry orchestration

Software Engineering Intern June 2026 – Aug. 2026
Pololu Robotics *Las Vegas, NV*

- Engineered an automated production test system in Arduino with serial CLI and fault handling, validating dual H-bridge MOSFET output, current thresholds, and fault flags across 15+ ADC channels, cutting QA cycle time by 10%
- Designed a robot-arm-compatible DUT fixture for a dual motor driver carrier with current sensing, relay-multiplexed load switching, and mixed-decay PWM verification; modeled enclosure in SolidWorks
- Refactored 500+ line legacy embedded codebase, resolved a critical timing race condition in PWM chopping logic, and validated power-on-reset sequencing for long-term maintainability

Software Engineering Intern June 2024 – Aug. 2024
Summer Business Institute Program (SBI) *Las Vegas, NV*

- Improved system uptime by resolving 25+ infrastructure and endpoint issues across Police, Fire, and Public Works departments
- Automated 3 cross-departmental reporting workflows by scripting data cleanup and consolidation pipelines, cutting weekly manual data entry by ~5 hrs

Research

Team Lead – Equine Airway Fluid Mechanics Aug. 2025 – Dec. 2025
Vertically Integrated Projects, Purdue University *West Lafayette, IN*

- Led a cross-disciplinary team in designing a particle image velocimetry (PIV) experiment to visualize airflow and particle deposition in a life-sized PDMS negative-mold equine airway phantom for studying recurrent airway obstruction (RAO)
- Conducted fluid dynamics simulations to extract velocity, pressure, and turbulence data, performing mesh independence studies and generating streamline/contour plots for validation against experimental PIV results

Projects

Orion | *Swift, Vision Framework, SSE* Apr. 2026 – Present

- Building a native macOS AI assistant with voice control, real-time gesture recognition via Apple's Vision framework, and full system automation through CGEvent
- Implemented streaming LLM responses via SSE with sentence-chunked TTS, achieving sub-1.5s end-to-end voice response latency

MCP Task Manager | *TypeScript, SQLite, Express, MCP Protocol* Jan. 2026 – Mar. 2026

- Engineered a dual-architecture task management system exposing 5 tools over MCP via stdio transport with an Express web dashboard backed by a shared SQLite database
- Designed a graceful degradation layer falling back from OpenAI API to a local regex-based parser with fuzzy typo tolerance, ensuring zero downtime during API failures

Honors & Awards

Claude Builder Hackathon: 1st place in the healthcare track for VEDA
Databricks Generative AI Fundamentals
Deloitte Data Analytics Certificate

Technical Skills

Languages: Java, Python, C, Swift, TypeScript, HTML/CSS
Tools & Platforms: Git, Arduino, SolidWorks, ONNX Runtime, ChromaDB, Jupyter Notebook, Tableau
Libraries & Frameworks: Matplotlib, Pandas, Express, Whisper, LLaVA, Vision Framework
Concepts: Object-Oriented Programming, Databases, Simulations, Data Analysis, Embedded Systems, RAG Pipelines